EXPERIMENT 4

Write a Python Program for the Crypt-Arithmetic Problem

Aim

To develop a Python program to solve a **Crypt-Arithmetic Problem** using **Backtracking Search**, where each letter represents a unique digit and the arithmetic equation is satisfied.

Objective

- To understand Constraint Satisfaction Problems (CSP) in Artificial Intelligence.
- To apply the Backtracking algorithm to solve Crypt-Arithmetic problems.
- To assign unique digits to letters while satisfying arithmetic constraints.
- To ensure that no two letters have the same digit and leading letters are not assigned zero.

Theory

A **Crypt-Arithmetic Problem** is a mathematical puzzle in which digits are replaced by letters. Each letter represents a unique digit from **0 to 9**, and the objective is to find the correct digit assignment that satisfies the given arithmetic equation.

A famous example is:

```
  SEND
+ MORE
------
 MONEY
```

Where:

- Each letter represents a unique digit.
- No two letters can have the same digit.
- The first letter of a number cannot be zero.

Since the number of possible combinations is very large, **Backtracking Search** is used to systematically assign digits and eliminate invalid assignments.

Algorithm

Step 1: Define the crypt-arithmetic equation.

Step 2: Identify all unique letters.

Step 3: Generate possible digit assignments (0–9).

Step 4: Assign digits to letters one by one.

Step 5: Check whether:

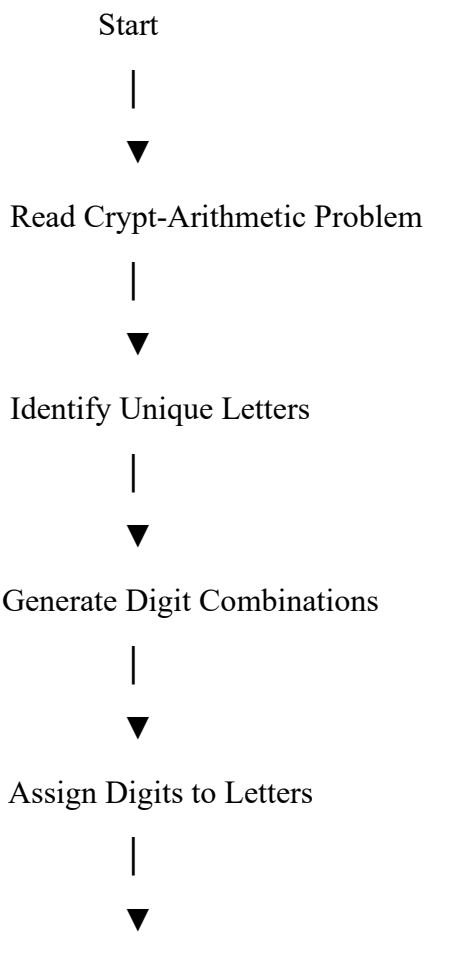
- Each letter has a unique digit.
- Leading letters are not assigned zero.
- The arithmetic equation is satisfied.

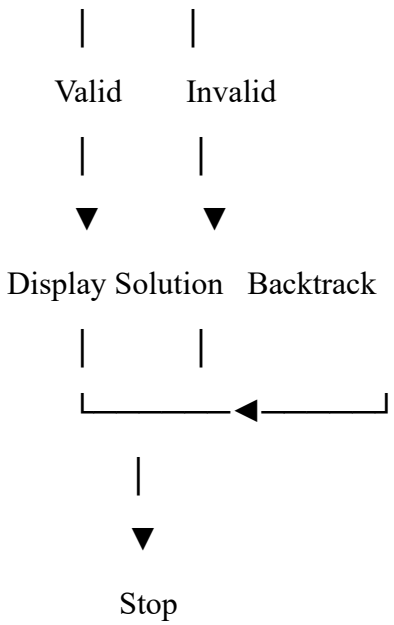
Step 6: If the assignment is valid, display the solution.

Step 7: Otherwise, backtrack and try another digit assignment.

Step 8: Continue until a valid solution is found.

Flowchart





Python Program

```
from itertools import permutations

letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')

for p in permutations(range(10), len(letters)):

    S, E, N, D, M, O, R, Y = p

    if S == 0 or M == 0:
        continue

    SEND = 1000*S + 100*E + 10*N + D
    MORE = 1000*M + 100*O + 10*R + E
    MONEY = 10000*M + 1000*O + 100*N + 10*E + Y

    if SEND + MORE == MONEY:
```

```
        print("Solution Found\n")

        print("SEND =", SEND)
        print("MORE =", MORE)
        print("MONEY =", MONEY)

        print("\nLetter Assignments")
        print("S =", S)
        print("E =", E)
        print("N =", N)
        print("D =", D)
        print("M =", M)
        print("O =", O)
        print("R =", R)
        print("Y =", Y)

        break
```

Sample Output

```
Solution Found

SEND = 9567
MORE = 1085
MONEY = 10652

Letter Assignments

S = 9
E = 5
```

N = 6

D = 7

M = 1

O = 0

R = 8

Y = 2

Explanation of Output

The program assigns unique digits to each letter:

Letter	Digit
S	9
E	5
N	6
D	7
M	1
O	0
R	8
Y	2

Verification:

```
  9567
+ 1085
-------
 10652
```

Since the equation is satisfied, the solution is correct.

Advantages

- Finds a valid solution systematically.
- Eliminates invalid assignments through backtracking.

- Ensures all constraints are satisfied.
- Suitable for solving Constraint Satisfaction Problems.

Applications

- Artificial Intelligence.
- Constraint Satisfaction Problems (CSP).
- Puzzle solving.
- Scheduling and planning.
- Logic-based reasoning systems.

Result

The Python program successfully solved the **Crypt-Arithmetic Problem** using the **Backtracking Algorithm** by assigning unique digits to letters while satisfying all arithmetic and uniqueness constraints.

Conclusion

The **Crypt-Arithmetic Problem** is a classic **Constraint Satisfaction Problem (CSP)** in Artificial Intelligence. By applying the **Backtracking Algorithm**, the program efficiently searches through possible digit assignments and eliminates invalid combinations until a valid solution is found. This experiment demonstrates the effectiveness of backtracking in solving complex logical and combinatorial problems while satisfying multiple constraints simultaneously.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Varshini M\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\Varshi
i M/OneDrive/Desktop/Here!!!/ARTIFICIAL INTELLIGENCE/VSC EXPs/EXP1.py"

SEND = 9567
MORE = 1085
MONEY = 10652

Letter Assignments
S = 9
E = 5
N = 6
D = 7
M = 1
O = 0
R = 8
Y = 2
```

C: > Users > Varshini M > OneDrive > Desktop > Here!!! > ARTIFICIAL INTELLIGENCE > VSC EXI

```python
from itertools import permutations

letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')

for p in permutations(range(10), len(letters)):

    S, E, N, D, M, O, R, Y = p

    if S == 0 or M == 0:
        continue

    SEND = 1000*S + 100*E + 10*N + D
    MORE = 1000*M + 100*O + 10*R + E
    MONEY = 10000*M + 1000*O + 100*N + 10*E + Y

    if SEND + MORE == MONEY:

        print("Solution Found\n")

        print("SEND =", SEND)
        print("MORE =", MORE)
        print("MONEY =", MONEY)

        print("\nLetter Assignments")
        print("S =", S)
        print("E =", E)
        print("N =", N)
        print("D =", D)
        print("M =", M)
        print("O =", O)
        print("R =", R)
        print("Y =", Y)

        break

```